

Deliverable 2 - Implemented Features

Final backend feature inventory and optional future enhancements

Account management and security

- Database-backed student registration with required-field, email, password, duplicate-email, and security-question validation.
- bcrypt password and recovery-answer hashing; plain-text passwords are never stored.
- Session-based login and logout with student/admin role separation.
- Server-protected student and admin pages plus protected API endpoints.
- Student profile retrieval and updates limited to the logged-in account.
- Security-question password recovery with one-time reset authorization.

Student event features

- Database-backed event list, event details, organizer information, registration totals, capacity, and remaining seats.
- Search across title, description, location, and organizer plus category, date, and status filters.
- Server-side registration rules for availability, date, capacity, duplicates, and event status.
- Personal My Registrations view with upcoming/past filters and ownership-protected cancellation.
- Database-driven dashboard summaries, upcoming registrations, category participation, progress, and suggested open events.
- Automatic Full status when the final seat is taken and automatic reopening when a registration is cancelled.

Admin event and attendance features

- Create, edit, cancel, reopen, disable, enable, complete, and delete events.
- Server-side validation of title, category, date, times, location, capacity, and status.
- Capacity cannot be reduced below the number of active registrations.
- Admin view of registered students for every event.
- Attendance marking as Attended or Missed while cancelled records remain protected.
- Database-calculated event totals, registrations, full-capacity events, attendance counts and rates, seat-fill percentage, registrations by category, and most popular category.

Architecture and quality

- Node.js and Express backend with modular routes, controllers, models, middleware, views, and public assets.
- SQLite persistence with foreign keys, uniqueness rules, database initialization, migration support, and repeat-safe demonstration data.
- Consistent responsive student/admin interfaces with shared navigation and footer components.

- JSON 404 responses for unknown API endpoints and centralized server error handling.
- Automated coverage for authentication, authorization, profiles, recovery, student registration ownership, duplicate/full/past/cancelled rules, admin CRUD, attendance, and statistics.

Optional future enhancements

Deliverable 2 contains the required core system. The following items are optional extensions rather than unfinished required features:

- Email reminders and calendar export for upcoming registrations.
- Event poster upload, QR-code attendance, and waitlists.
- A production session store and deployment configuration.
- Additional automated browser tests and richer charts.